

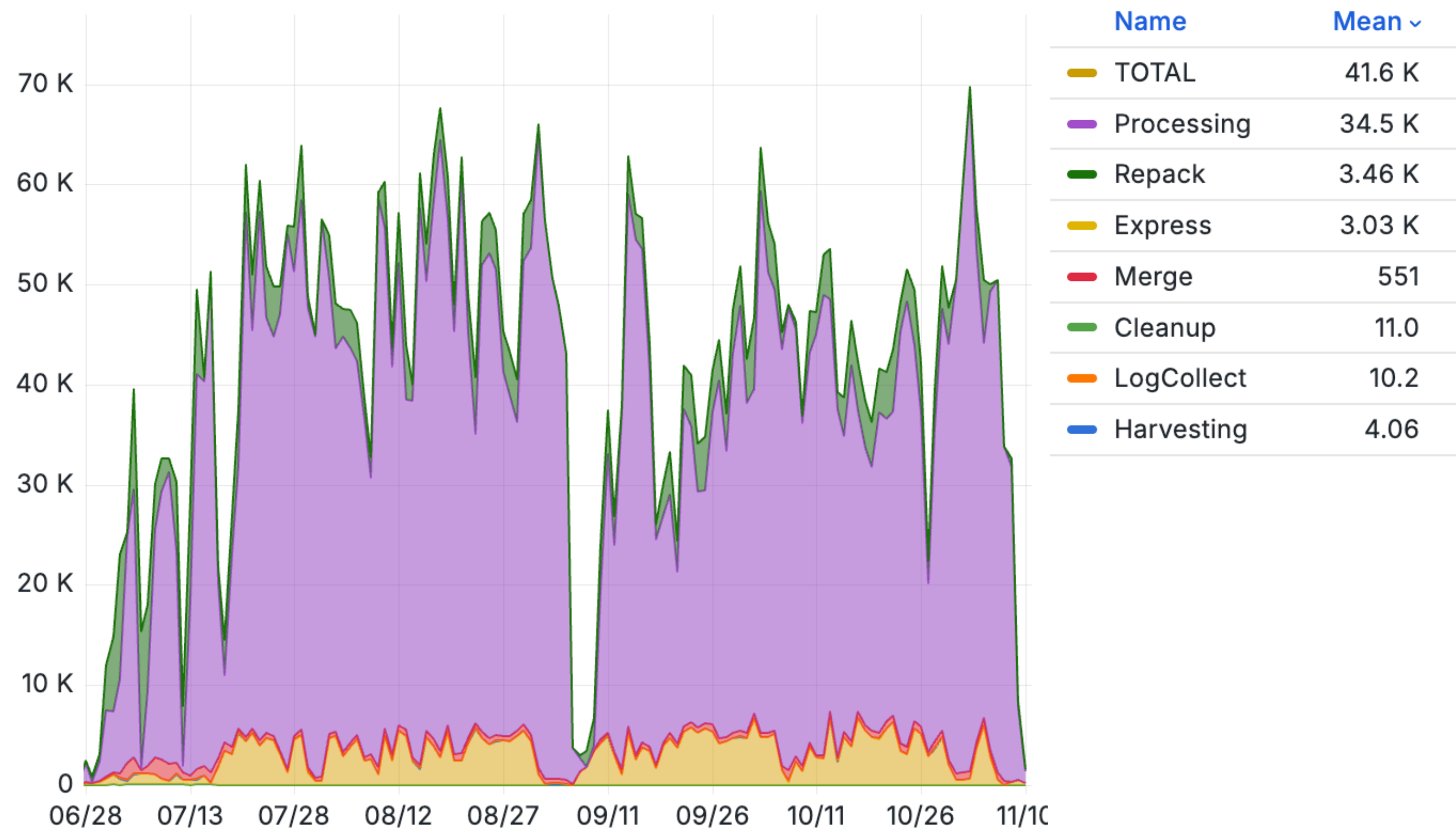
Tier-0 evolution for Run 4

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Tier-0 in LS3

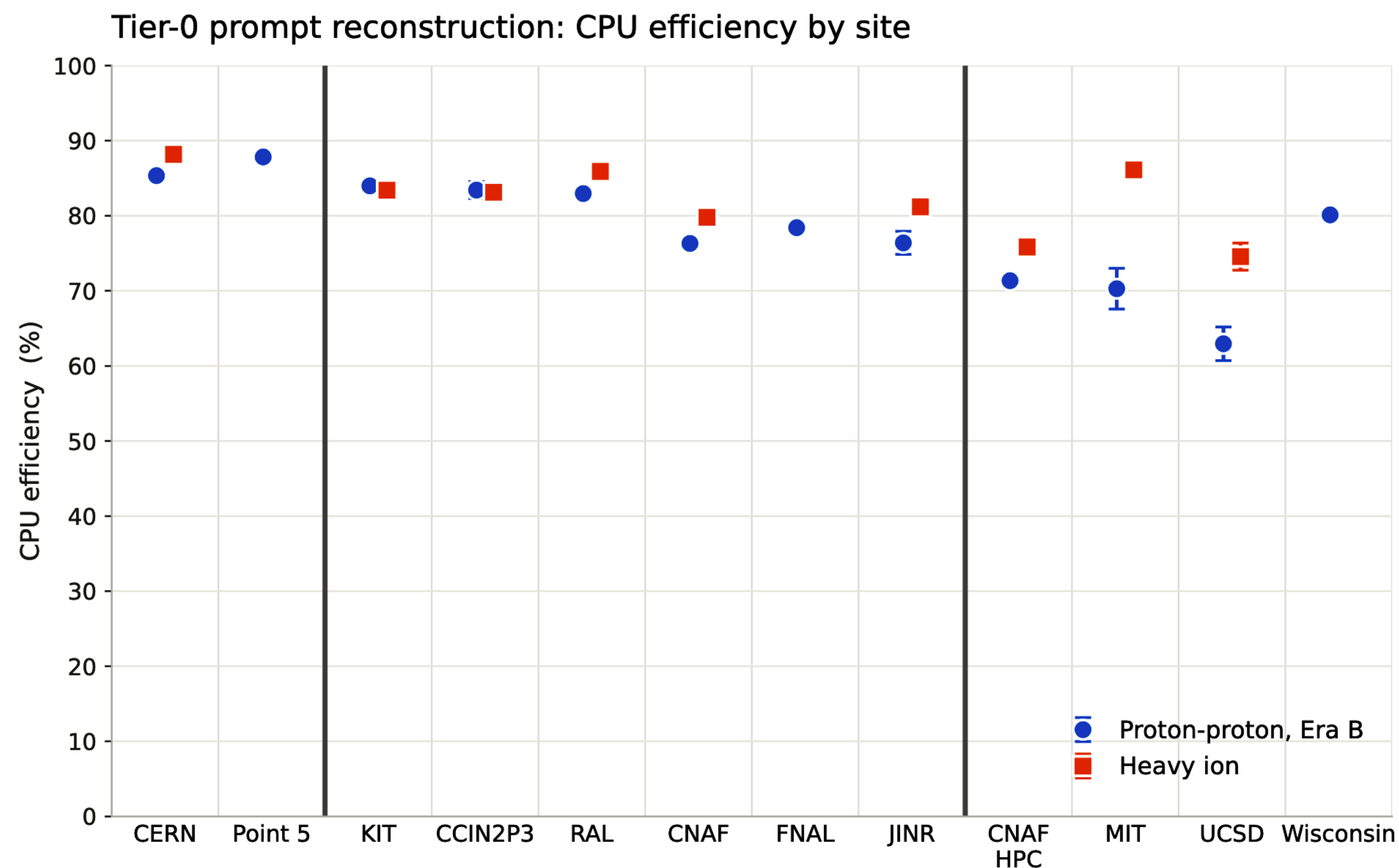
- From 2027 there won't be any support for Tier-0 operations or development until around 2029
 - CERN operator position is reallocated to the development of the new workflow management system (WMS)
- First working setup of the new WMS is expected in 2028
 - Monte Carlo production is the primary focus
- Tier-0 is effectively a specialized workflow in any WMS
 - Primary focus is reliable and timely data processing
 - Data repacking and distribution
 - Express processing for data calibration
 - Prompt reconstructions

Running CPUs at CERN



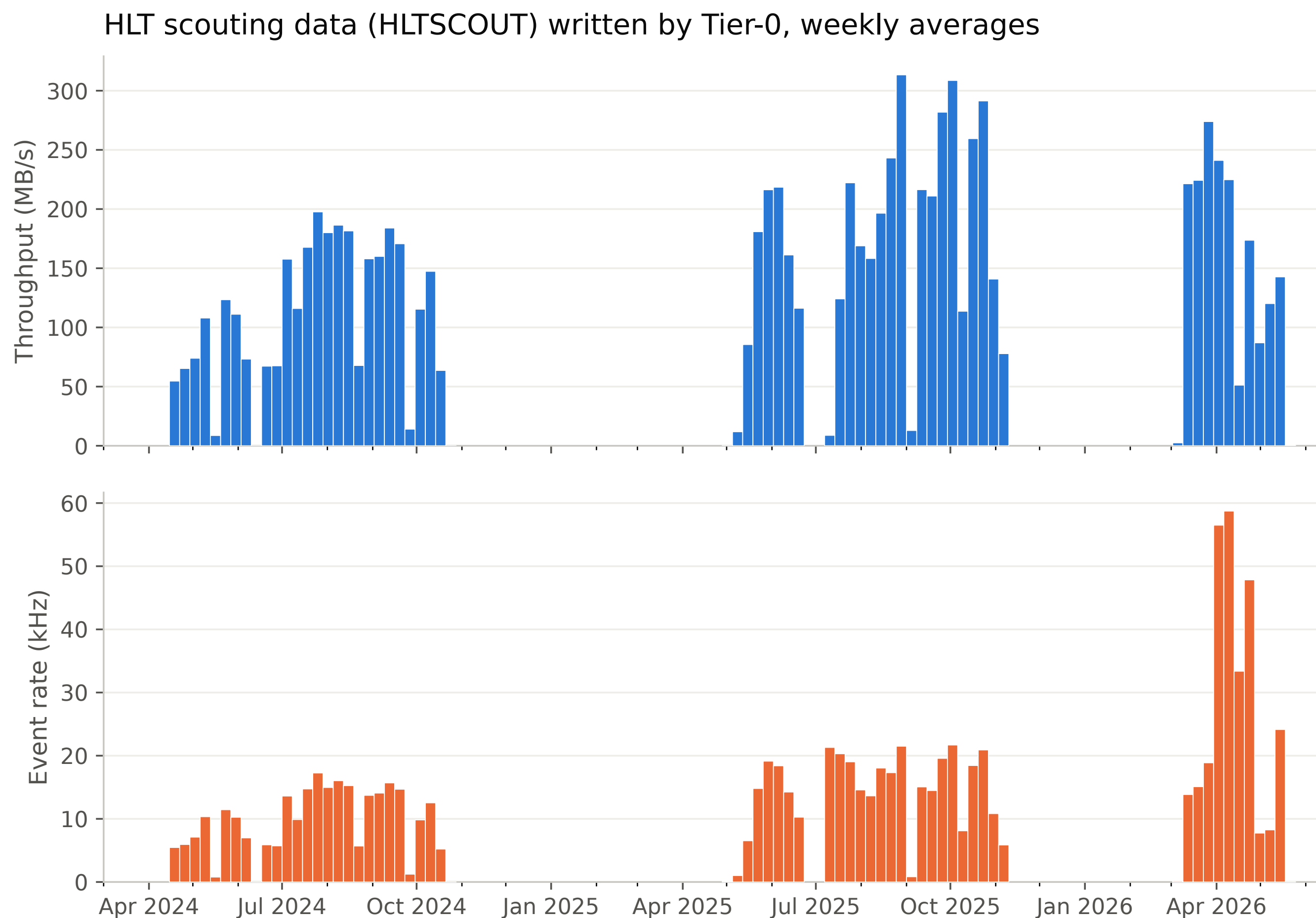
- Run3 Express CPU usage at CERN: ~ 7%
- Prompt reco delay requires disk space
 - Disk space is expensive and will be tight
 - Each additional day in Run3 ~ 1PB of extra space
 - Scale it by projected increase in data rates (at least 2x)
 - 48h is realistic
 - 7 days is probably doable
- We need to rely on rolling calibrations
 - Can use more CPU, but not disk
- Calibration workflows must have strict time limits

Prompt Reconstruction



- CPU is also a limited resource
 - CERN + EU Tier-I is all that we can realistically rely on
 - Other sites are less efficient to read data from CERN
- Most likely some data will be parked
 - Expect significant delays in getting it reconstructed

Run4 Scouting



- Post-processing of HLT and L1 Scouting is a way to reduce CPU demand for HLT
 - Object ID, B-tagging etc
 - May include JES and other calibrations if available
- Peak scouting event rates in Run3
 - pp: 20 kHz
 - Low PU: 60 kHz
- With 750 kHz in Run 4 it will be a major load
 - Likely only CERN can handle such load

- Most important output for data analysis
 - Critical for POG and DPGs as well
- We have CPU capacity to produce it all, but not as independent workflows
 - Operational load growth proportionally to the number of workflow to handle
- Central production may be able to handle it instead, but it will come with delays